

MUST: The First Dataset and Unified Framework for Multispectral UAV Single Object Tracking

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Abstract

UAV tracking faces significant challenges in real-world scenarios, such as small-size targets and occlusions, which limit the performance of RGB-based trackers. Multispectral images (MSI), which capture additional spectral information, offer a promising solution to these challenges. However, progress in this field has been hindered by the lack of relevant datasets. To address this gap, we introduce the first large-scale Multispectral UAV Single Object Tracking dataset (MUST), which includes 250 video sequences spanning diverse environments and challenges, providing a comprehensive data foundation for multispectral UAV tracking. We also propose a novel tracking framework, UNTrack, which encodes unified spectral, spatial, and temporal features from spectrum prompts, initial templates, and sequential searches. UNTrack employs an asymmetric transformer with a spectral background eliminate mechanism for optimal relationship modeling and an encoder that continuously updates the spectrum prompt to refine tracking, improving both accuracy and efficiency. Extensive experiments show that our proposed UNTrack outperforms state-of-the-art UAV trackers. We believe our dataset and framework will drive future research in this area. The dataset is available on <https://github.com/q2479036243/MUST-Multispectral-UAV-Single-Object-Tracking>.

1. Introduction

UAV tracking is critical but often encounters complex challenges, which hinder the extraction of accurate spatial features, as shown in Fig. 1. These factors substantially degrade the performance of RGB-based trackers that rely primarily on visual cues [13, 14]. Furthermore, the target's appearance may change significantly during tracking, often deviating considerably from the initial template, thereby making the tracker prone to drift.

Multispectral images provide additional spectral infor-

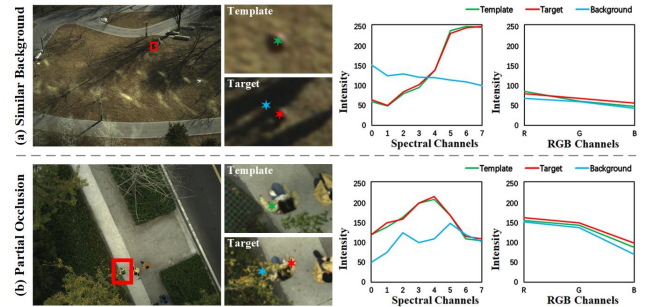


Figure 1. In challenging scenarios, the spatial features (e.g., color and texture) of the tracked target closely resemble those of the background, making differentiation and localization difficult. However, the target's spectral information differs significantly from the background and aligns with the template's spectral data, providing robust features for reliable tracking.

mation that characterizes the target's intrinsic reflectance properties [27], offering valuable insights in situations where spatial features are limited by the aforementioned challenges. For example, even with severe background interference and color similarities, distinct spectral curves allow for effective target differentiation and localization (Fig. 1). Moreover, key spectral features tend to remain stable during tracking, improving robustness. Thus, multispectral imaging is a promising solution for UAV tracking in challenging scenarios. However, the lack of large-scale, challenging datasets hampers progress in this area.

To address this gap, we present the first large-scale Multispectral UAV Single Object Tracking dataset (MUST). The dataset comprises 250 video sequences, totaling 43,000 frames, each with 1200×900 pixels and 8 spectral bands that encompass both visible-light and near-infrared bands. Furthermore, the dataset incorporates 12 key challenge attributes, including small-size targets, background clutter, and similar colors, making it highly representative of real-world challenges. By leveraging data diversity and including difficult scenarios, the MUST dataset provides crucial support for the development and thorough evaluation of

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multispectral UAV tracking methods.

In addition, we introduce a novel tracking framework, termed the Unified Spectral-Spatial-Temporal Tracker (UNTrack), designed specifically for multispectral UAV tracking. UNTrack simultaneously integrates spectral, spatial, and temporal information in a unified manner by incorporating the historical spectrum prompt, the initial template, and sequential searches from consecutive frames. To better extract collaborative features from the spectral, spatial, and temporal domains, UNTrack employs an asymmetric transformer that models the core relationships among various feature domains while pruning attention maps irrelevant to tracking. This innovative asymmetric structure streamlines and compacts joint feature extraction, striking an optimal balance between accuracy and efficiency.

To further reduce noisy interference and computational costs, UNTrack employs a background suppression mechanism that efficiently eliminates background regions from the search image based on spectral characteristics. Additionally, UNTrack incorporates an encoder that generates a spectral prompt that captures the unique spectral features of the target, enabling effective discrimination between the target and similar background. The encoded spectral prompt is then used to update the input for subsequent tracking, providing stable spectral information that is free from external interference, thereby enhancing tracking robustness.

We conducted a thorough evaluation of the proposed UNTrack and existing trackers using our newly introduced MUST dataset. UNTrack demonstrates state-of-the-art performance, and extended experiments on other multispectral or general object tracking datasets highlight its superiority and versatility. In summary, the main contributions of this work are as follows:

- We present MUST, the first large-scale and challenging multispectral UAV tracking dataset, providing foundational data support to advance research in this field.
- We propose a novel framework, UNTrack, that integrates spectral, spatial, and temporal information in a unified manner for robust tracking in challenging scenarios.
- We introduce an asymmetric transformer that models the core relationships among various feature domains, pruning attention maps and improving tracking efficiency.

2. Related Work

UAV Tracking Datasets. With the growing interest in UAV tracking, numerous related datasets have emerged. Examples include Stanford Drone [29], UAV123 [26], and others [7, 15, 41], which provide diverse data to the field. However, these RGB-based datasets struggle with the challenges of obvious appearance variations in applications, limiting their effectiveness in evolving tracking needs. To address this limitation, Xiong *et al.* [34] introduced the first MSI-based tracking dataset, but it focuses on general objects

thereby unsuitable for UAV tracking. To our knowledge, there are currently no multispectral datasets specifically designed for UAV tracking. In response, we propose a large-scale dataset with a wide range of challenging attributes, addressing the issue of data scarcity.

Multispectral UAV Trackers. In general visual tracking, one-stream trackers [36] integrate feature extraction and interaction within a single framework, offering superior performance compared to Siamese trackers [11] and becoming the mainstream approach. However, existing UAV trackers are still primarily based on the Siamese architecture [16], while only focusing on RGB data and ignoring more potential multispectral images. Furthermore, previous studies [19, 20] mostly adapted RGB-based trackers to handle multispectral images without fully exploiting spectral features. To address these issues, we propose the first multispectral UAV tracker, equipped with a novel tracking framework, to tackle challenges based on unified features.

Prompt Learning. Recent studies have introduced prompts to improve tracking performance [1, 40], but these approaches focus primarily on appearance, overlooking the more fundamental spectral characteristics. Moreover, previous trackers [5, 30] have not explored the optimal attention structure for integrating prompts. The unified extraction of spectral, spatial, and temporal features remains a critical challenge. To address this, we propose an asymmetric transformer and a spectrum prompt encoder that strike a balance between tracking accuracy and inference efficiency.

3. MUST Dataset

3.1. Overview

Data Collection. The proposed Multispectral UAV Single Object Tracking (MUST) dataset is captured using a professional-grade UAV equipped with a multispectral camera. To enhance data diversity, we collected data across various scenarios and time periods, covering 250 video sequences, of which 160 are used for training and 90 are used for testing. Each multispectral image frame has a spatial resolution of 1200×900 and 8 spectral bands. A detailed description of the dataset parameters and processing strategies is provided in the supplementary materials.

Dataset Comparison. Table 1 shows the comparison of our proposed MUST dataset against other UAV or multispectral tracking datasets. Although RGB-based datasets [15, 26] are well-established, their single data modality limits the use of spectral information. Furthermore, the existing MSI-based tracking dataset [34] is limited by its small-scale and artificial ground scenarios, restricting its applicability to general tracking. In contrast, our proposed MUST dataset offers a larger number of video sequences with extended durations, ensuring both data diversity and broad challenges. In addition, we collected UAV data across a broader range

Table 1. Comparison with other UAV perspective datasets and MSI-based datasets. – means no relevant information was provided.

Dataset	Modality	Task	Sequences	Frames	Resolution	Channels	Durations	Altitude	Year
UAV123 [26]	RGB	UAV SOT	123	113 k	1280×720	3	3800 s	5~25 m	2016
DTB70 [15]	RGB	UAV SOT	70	16 k	1280×720	3	640 s	0~120 m	2017
VisDrone [41]	RGB	UAV SOT	167	139 k	Not Uniform	3	4600 s	–	2021
VTUAV [37]	RGB-T	UAV SOT	500	1700 k	1920×1080	4	55000 s	5~20 m	2022
HOT [34]	MSI	General SOT	50	21 k	512×256	16	840 s	–	2020
MUST (Ours)	MSI	UAV SOT	250	43 k	1200×900	8	8600 s	20~250 m	2024

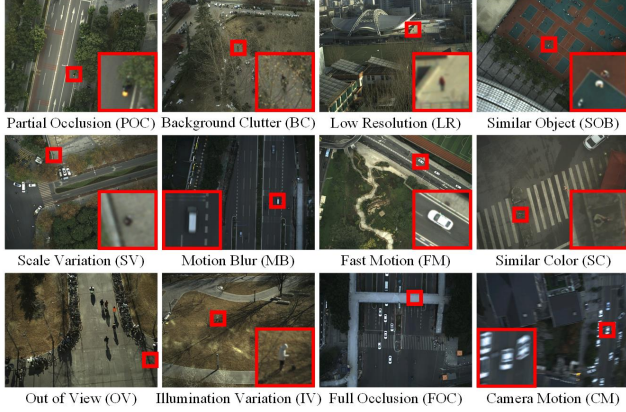


Figure 2. Examples of representative challenge scenarios, with targets marked by red boxes and displayed in magnified views.

of flight altitudes with various flight postures, making it more reflective of real-world scenarios.

3.2. Dataset Analysis

Challenge Attributes. To reflect the critical aspects of UAV tracking, we have defined 12 key challenge attributes as illustrated in Fig. 2. These attributes offer a comprehensive overview of UAV tracking, covering target position, background characteristics, and UAV status. These challenges also highlight the necessity of using spectral information for UAV tracking. A more detailed explanation of these challenges is provided in the supplementary material.

Statistical Analysis. We visualize the statistical properties of the MUST dataset to provide a comprehensive analysis. As shown in Fig. 3 (a), the 12 key challenge attributes are evenly distributed, ensuring the difficulty and diversity of both subsets. Moreover, we count the target scale distribution, as depicted in Fig. 3 (b). Most targets are small-sized, with their sizes accounting for around 3×10^{-5} of the entire image. The prevalence of small targets is due to the shooting altitude up to 200 meters, which brings new challenges.

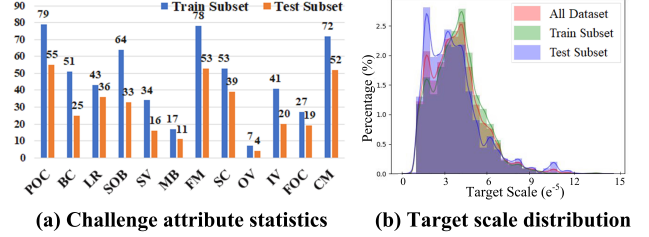


Figure 3. Statistics of the MUST dataset: (a) Distribution of video sequences across 12 key challenges, (b) Target scale distribution and consistency across different subsets.

4. Method

We propose the Unified Spectral-Spatial-Temporal Tracker (UNTrack) as a baseline for multispectral UAV tracking, as shown in Fig. 4. UNTrack utilizes a unified asymmetric transformer to extract features from spectral, spatial, and temporal domains, while suppressing background noise in search regions. The pruned attention maps are fed into a spectrum prompt encoder, which generates prompts to deliver historical spectral characteristics. Finally, the prediction head updates the spectral-based prompt and outputs the target’s position in the current frame.

4.1. Unified Asymmetric Transformer

Revisit Transformer. The vanilla transformer [31] is commonly employed for feature extraction in one-stream trackers due to its exceptional long-range modeling ability. The input template and search are fed into a patch embedding layer and concatenated to form tokens F . Subsequently, F is linearly projected to obtain query Q , key K , and value V , with their respective linear projection weights W_Q , W_K , and W_V . The attention map A is computed as follows:

$$Q = FW_Q, K = FW_K, V = FW_V, \quad (1)$$

$$A = \text{Attn}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (2)$$

where $\sqrt{d_k}$ is a scaling factor.

Unified Embedding. The vanilla transformer focuses on spatial features but neglects spectral and temporal information. In contrast, the proposed UNTrack incorporates both

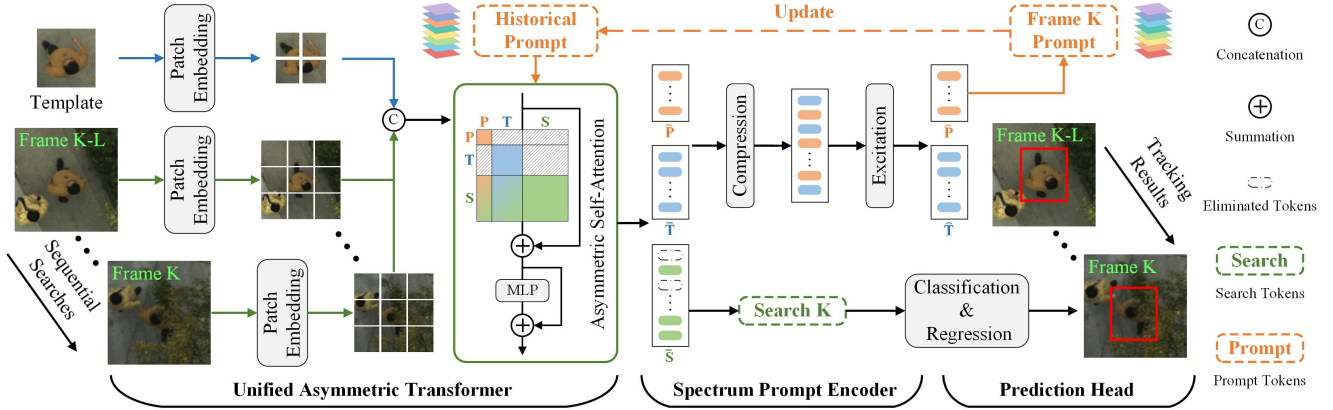


Figure 4. The proposed UNTrack consists of three components: the unified asymmetric transformer, the spectrum prompt encoder, and the prediction head. UNTrack takes the spectrum prompt, initial template, and sequential search as unified inputs and outputs the target bounding box for each frame. The encoded prompt tokens of the current frame update the spectrum prompt for subsequent tracking.

sequential and historical input, enabling unified spectral-spatial-temporal feature extraction and relation modeling. Specifically, UNTrack initially treats consecutive frames as sequential searches $\mathbf{I}_S = \{\mathbf{I}_{S_i} \in \mathbb{R}^{H_S \times W_S \times 8} | i = 1, \dots, N\}$ and embeds them together with the initial template $\mathbf{I}_T \in \mathbb{R}^{H_T \times W_T \times 8}$. Here, $\mathbf{I}_{S_i}$ denotes each search frame, and N is the total frame number. The process is expressed as:

$$\mathbf{T} = \phi(\mathbf{I}_T) \in \mathbb{R}^{L_T \times C}, \quad (3)$$

$$\mathbf{S} = [\phi(\mathbf{I}_{S_1}); \dots; \phi(\mathbf{I}_{S_N})] \in \mathbb{R}^{L_S \times C}, \quad (4)$$

where $\mathbf{T}$ and $L_T = \frac{H_T \times W_T}{16}$ denote the template tokens and their length. $\mathbf{S}$ and $L_S = \frac{N \times H_S \times W_S}{16}$ represent the concatenated search tokens and their total length. C is the feature dimension of the tokens, and ϕ denotes the separation and flattening operations. Furthermore, UNTrack concatenates the historical prompt tokens $\mathbf{P} \in \mathbb{R}^{L_P \times C}$ from the previous frame, where L_P represents the token length, typically set to 1 in practice. The resulting $\mathbf{F} = [\mathbf{P}; \mathbf{T}; \mathbf{S}]$ is then used for subsequent attention calculations.

Asymmetric Attention. Different from previous trackers, UNTrack adopts the unified tokens $\mathbf{F}$, resulting in a modified attention calculation formula, as follows:

$$\mathbf{Q} = [\mathbf{Q}_P; \mathbf{Q}_T; \mathbf{Q}_S], \mathbf{K} = [\mathbf{K}_P; \mathbf{K}_T; \mathbf{K}_S], \mathbf{V} = [\mathbf{V}_P; \mathbf{V}_T; \mathbf{V}_S], \quad (5)$$

$$\mathbf{A} = \text{Softmax}\left(\frac{[\mathbf{Q}_P; \mathbf{Q}_T; \mathbf{Q}_S][\mathbf{K}_P; \mathbf{K}_T; \mathbf{K}_S]^T}{\sqrt{d_k}}\right)[\mathbf{V}_P; \mathbf{V}_T; \mathbf{V}_S]. \quad (6)$$

According to the information streams pipeline, the calculated attention map $\mathbf{A}$ is divided into nine blocks. As depicted in Fig. 5 (a), each block corresponds to the attention results of different token interactions, and their detailed descriptions are offered in the supplementary material.

Notably, each block is unevenly relevant to the tracking process. For instance, blocks 1, 5, and 9 denote the self-attention of tokens, extracting deep semantic features and representing target characteristics. Blocks 7 and 8 capture

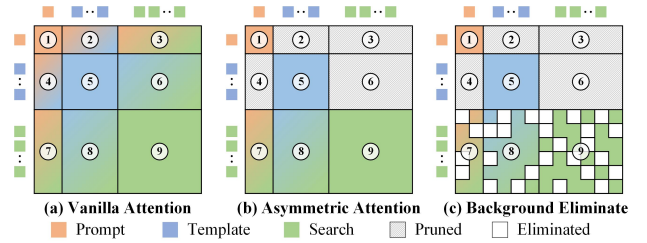


Figure 5. Information streams in the attention mechanism. The interaction between prompt, template, and search corresponds to the nine blocks in the attention map.

the cross-attention between the search and target, which is responsible for locating essential search regions. Conversely, the remaining blocks focus on the prompt and template, which are not only irrelevant to the tracking process but also brings computational cost and noise interference.

Inspired by the above insights, we propose an asymmetric attention mechanism, as shown in Fig. 5 (b). Specifically, we prune the cross-attention on the prompt and template, and the attention map can be calculated as:

$$\mathbf{A} = [\text{Attn}(\mathbf{Q}_P, \mathbf{K}_P, \mathbf{V}_P), \text{Attn}(\mathbf{Q}_T, \mathbf{K}_T, \mathbf{V}_T), \text{Attn}(\mathbf{Q}_S, \mathbf{K}, \mathbf{V})]. \quad (7)$$

The proposed asymmetric structure preserves necessary attention maps, facilitating spectral-spatial-temporal features extraction at low cost.

Spectral Background Eliminate. Based on asymmetric attention, UNTrack introduces a spectral background eliminate mechanism as shown in Fig. 5 (c). The mechanism aims to progressively eliminate search regions belonging to the background, thereby reducing computational overhead and enhancing focus on the target. Specifically, the attention map $\mathbf{A}_S \in \mathbb{R}^{(L_P + L_T + L_S) \times C}$ for the search tokens de-

scribes the correlation of each search region with the target from different perspectives.

This provides a feasibility to quantify the confidence of each search region belonging to the target by calculating a scalar value, which is given by:

$$B = \|\text{AvgPool}(\mathbf{A}_S)\|, \quad (8)$$

where AvgPool refers to the average pooling along the token length, and $\|\cdot\|$ denotes the Euclidean norm along the spectral dimension. A region is more likely to be background if its value is relatively small. Therefore, we keep the top $\rho \times (L_P + L_T + L_S)$ regions as candidates, while eliminating the remaining search regions. Here, ρ indicates the keeping ratio, which is dynamically adjusted using a cosine annealing strategy during training to progressively eliminate background.

4.2. Spectrum Prompt Encoder

In multispectral UAV tracking, spectral information facilitates background discrimination and plays a pivotal role in addressing various challenges. To take advantage of this, UNTrack adopts a spectral prompt encoder, as illustrated in Fig. 4. The encoder takes the unified tokens $\bar{\mathbf{F}} = [\bar{\mathbf{P}}; \bar{\mathbf{T}}; \bar{\mathbf{S}}]$ as input, which are obtained from the asymmetric transformer. Here, $\bar{\mathbf{P}}$, $\bar{\mathbf{T}}$, and $\bar{\mathbf{S}}$ correspond to the generated prompt, template, and search tokens, respectively. The prompt and template tokens are then concatenated to generate $[\bar{\mathbf{P}}; \text{GPool}(\bar{\mathbf{T}})] \in \mathbb{R}^{L_P \times 2C}$ where GPool denotes global pooling along the token length, which highlights the target's spectral features.

Subsequently, UNTrack employs CNN-based layers to process the spectral features of the prompt and template, enabling attention interaction. This process is described as:

$$\mathbf{A}' = \text{FC}([\bar{\mathbf{P}}; \text{GPool}(\bar{\mathbf{T}})]) \in \mathbb{R}^{L_P \times \frac{2C}{r}}, \quad (9)$$

$$\mathbf{A}'' = \text{FC}(\mathbf{A}') \in \mathbb{R}^{L_P \times 2C}, \quad (10)$$

$$[\hat{\mathbf{P}}; \hat{\mathbf{T}}] = \mathbf{A}'' + \text{MLP}(\mathbf{A}'') \in \mathbb{R}^{L_P \times 2C}, \quad (11)$$

where r denotes the compression ratio, and both FC layers are implemented via convolution. The first FC layer performs spectral compression, while the second FC layer is responsible for excitation. The compression-excitation operation enables spectral attention calculation for the prompt and template. The MLP denotes a multi-layer perceptron, which does not alter the spectral dimensions. The final encoded $\hat{\mathbf{P}} \in \mathbb{R}^{L_P \times C}$, recording the target's material properties, is regarded as the spectrum prompt for subsequent frame tracking.

4.3. Prediction Head and Loss Function

UNTrack adopts a dual-branch structure prediction head, which processes the search tokens $\bar{\mathbf{S}}$ as input. The head initially re-interprets tokens into 2D spatial feature maps and

then feeds them into two separate branches responsible for classification and regression. The classification branch outputs a confidence map $\text{Cla} \in \mathbb{R}^{1 \times H_S \times W_S}$, while the regression branch predicts a bounding box $\text{BBox} \in \mathbb{R}^{4 \times H_S \times W_S}$, both generated through stacked convolutional layers.

During training, we adopt focal loss [10] for classification, denoted as $\mathcal{L}_{\text{cls}}$. For bounding box regression, we employ both $\mathcal{L}_1$ loss and GIoU loss [28] (denoted as $\mathcal{L}_{\text{GIoU}}$). The total loss $\mathcal{L}$ for model convergence is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_{\text{GIoU}}, \quad (12)$$

where $\lambda_1 = 5$ and $\lambda_2 = 2$ are the weighting factors that balance the contribution of each loss term.

5. Experiment

5.1. Implementation Details

We train and evaluate numerous state-of-the-art trackers on the proposed Multispectral UAV Single Object Tracking (MUST) dataset, each with fine-tuned hyperparameters for optimal performance. Moreover, we evaluate our Unified Spectral-Spatial-Temporal Tracker (UNTrack) across multiple datasets. We adopt the AdamW optimizer with an initial learning rate of 10^{-4} , resizing searches to 384×384 and templates to 192×192 . We train for 50 epochs with a batch size of 24 and reduce the learning rate by a factor of 10 after 30 epochs. All experiments are conducted on NVIDIA RTX3090 GPUs.

Due to differences in spectral resolution and range across datasets, there are no available MSI-based pre-trained parameters. We thus load parameters from the ImageNet dataset [6] and adopt a simple yet effective parameter reconstruction strategy. The strategy aims to extend RGB-based pre-trained parameters to MSI-based vision tasks through interpolation. We provide a detailed description and analysis of this strategy in the supplementary material. Table 2 presents the performance of trackers with different parameter sources, with * marking the reconstructed parameters and otherwise loading RGB-based parameters directly.

5.2. Results on the MUST Dataset

Main Results. We compare UNTrack against other state-of-the-art trackers on the MUST dataset, as shown in Tab. 2. Obviously, one-stream trackers outperform Siamese trackers and have become the dominant approach in visual tracking tasks. Additionally, the parameter reconstruction strategy significantly improves the accuracy, with an average increase of over 10% across metrics, as confirmed on various trackers. Compared with other trackers, UNTrack achieves state-of-the-art performance, for example, surpassing OS-Track [36] by 4.6% in AUC and 5.9% in Precision (Pre).

Attribute-based Comparison. Furthermore, we compare the performance of various trackers on each challenge attribute, as illustrated in Fig. 6. Compared to other trackers,

Table 2. Comparison with state-of-the-art trackers on the MUST dataset. * denotes trackers trained with reconstructed parameters. The top two results are highlighted in red and blue.

Method	AUC	SR _{0.5}	SR _{0.75}	Pre	Pre _N
SiamRPN [11]	38.9	49.5	22.3	56.7	50.1
SiamMask [32]	39.1	51.0	25.3	56.1	52.8
DiMP [2]	47.3	60.7	31.6	69.8	57.1
TaMOs [25]	20.4	24.5	12.0	31.0	23.8
ToMP [24]	30.3	39.1	14.5	46.1	40.5
SGDViT [35]	38.1	50.2	22.8	54.4	52.1
OTrack ₂₅₆ [36]	39.9	50.5	27.4	55.9	51.2
OTrack ₃₈₄ [36]	44.5	56.7	27.1	63.9	56.8
ODTrack [40]	46.3	58.4	28.1	67.6	60.4
ZoomTrack [9]	44.0	56.2	25.2	63.5	55.9
OTrack ₂₅₆ * [36]	52.3	64.8	36.3	67.5	65.2
OTrack ₃₈₄ * [36]	55.1	69.6	44.1	73.3	68.8
ODTrack* [40]	52.7	67.3	38.8	71.3	66.2
ZoomTrack* [9]	53.3	67.7	40.4	72.3	66.7
UNTrack (Ours)	47.8	61.0	28.9	68.3	59.4
UNTrack* (Ours)	59.7	75.8	46.2	79.2	74.8

UNTrack achieves superior accuracy across all challenges. Notably, UNTrack excels under demanding motion conditions such as MB and CM. UNTrack’s robustness is primarily due to its emphasis on the spectrum, which remains undisturbed by violent motion, thereby reducing tracking drift. Additionally, UNTrack leverages unified spectral-spatial-temporal features, demonstrating exceptional potential in handling challenges with restricted appearance, such as SC and FOC. When handling the most challenging OV attribute, UNTrack employs historical prompt and sequential searches enabling successful re-tracking despite short-term target loss, and outperforming previous trackers.

Evaluation of state-of-the-art trackers on the MUST dataset demonstrates its feasibility as reliable data support. Notably, the defined 12 key challenge attributes emphasize the unique characteristics of multispectral UAV data, laying the groundwork for targeted improvements in trackers. Moreover, extensive experiments show the superior of UNTrack. In conclusion, we have established a pioneering benchmark for multispectral UAV tracking.

Visualization of Background Elimination. To intuitively demonstrate the superiority of UNTrack, we visualize its behavior for qualitative analysis. We first present the background elimination process at different stages in the asymmetric transformer, providing an in-depth analysis of the spectral background elimination mechanism, as shown in Fig. 7. UNTrack gradually eliminates search regions belonging to the background based on spectral confidence, greatly reducing computational overhead.

Visualization of Tracking Results. Furthermore, we compare the tracking performance of UNTrack against various

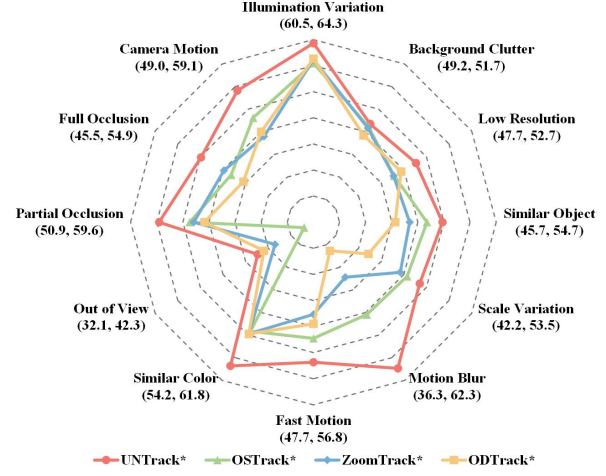


Figure 6. A comparison of UNTrack with other state-of-the-art trackers in terms of AUC across each challenge attribute on MUST dataset, with the lowest and highest performance values marked.

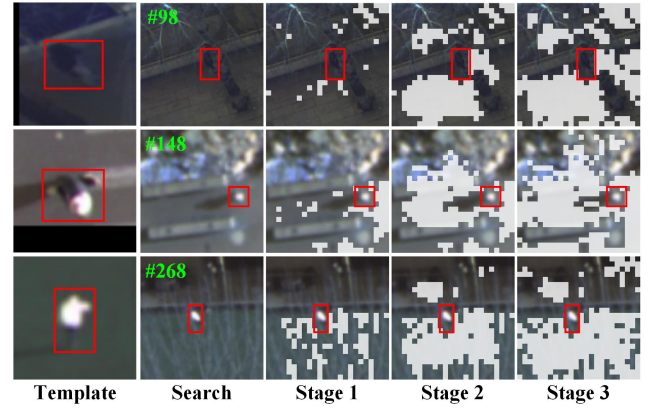


Figure 7. Visualization of the progressive background elimination process using the spectral background elimination mechanism. The target is highlighted with a red box, and the white masks indicate the eliminated regions.

trackers, as shown in Fig. 8. Despite the presence of objects with similar appearances or backgrounds of similar colors around the target, UNTrack consistently delivers robust tracking results, surpassing other trackers. In addition, attention heatmaps illustrate the focus regions of UNTrack in each search frame. Obviously, UNTrack maintains a steady focus on the target amidst various challenges, an ability enabled by leveraging spectral features.

We visualize the tracking results for various challenge attributes that are intractable for traditional RGB-based trackers. As shown in Fig. 9, RGB-based trackers struggle to distinguish the target when it is surrounded by similar objects or background. When the target is occluded, the template features extracted by RGB-based trackers are contaminated,

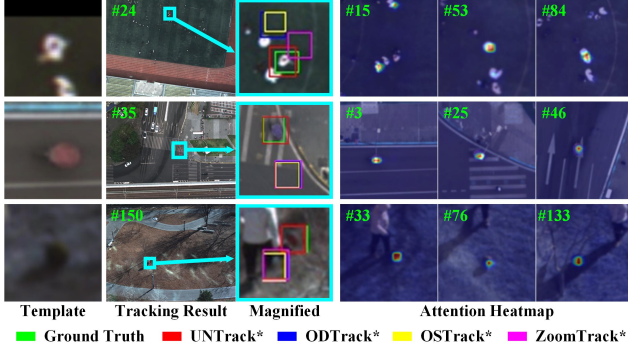


Figure 8. Comparison of tracking results for UNTrack and other state-of-the-art trackers, with the target area magnified. The attention heatmap highlights the regions of focus for UNTrack.

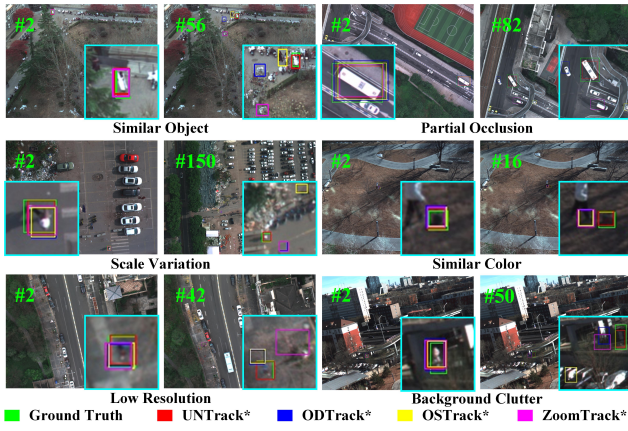


Figure 9. Comparison of UNTrack and other state-of-the-art trackers when dealing with various challenge attributes.

leading to tracking drift. In addition, RGB-based trackers cannot obtain sufficient spatial features from the small-sized target, thus performing poorly. In contrast, UNTrack adopts the novel spectral-spatial-temporal unified tracking framework, providing stable tracking results across all challenges.

5.3. Versatility Evaluation

MSI-Based General Tracking. The HOT [34] is an available dataset used for MSI-based general tracking. We conduct extensive comparative experiments based on it and additionally introduce Frames Per Second (FPS) as a metric for tracker efficiency. For a fair comparison, all trackers perform inference on single NVIDIA RTX 3090 GPU. As shown in Tab. 3, UNTrack achieves the fastest inference speed at 37 FPS while also reaching state-of-the-art tracking accuracy. This optimal trade-off between accuracy and efficiency is enabled by the asymmetric attention structure and spectral background eliminate mechanism, which effectively eliminate redundant tokens.

RGB-Based General Tracking. We extend UNTrack to

Table 3. Comparison of results for UNTrack and other state-of-the-art trackers on the HOT dataset, with the top two results highlighted in red and blue.

Method	AUC	Pre	FPS
MHT [34]	58.7	88.0	–
BAENet [17]	60.6	87.7	–
SSTNet [18]	62.1	90.1	–
SiamHYPER [22]	67.8	94.5	27.7
TMTNet [39]	69.9	92.8	12.6
SSTFT [33]	68.2	88.4	–
SEENet [19]	65.3	93.0	13.1
HANet [23]	68.8	94.8	21.2
UNTrack (Ours)	70.4	93.7	37.0

Table 4. The performance of our UNTrack and other state-of-the-art trackers on the GOT10K dataset. The best two results are highlighted in red and blue.

Method	AO	SR _{0.5}	SR _{0.75}
SiamRPN++ [12]	51.7	61.6	32.5
AutoMatch [38]	65.2	76.6	54.3
OTrack [36]	71.0	80.4	68.2
MixFormer [5]	72.5	82.4	69.9
CTTrack [30]	71.3	80.7	70.3
SeqTrack [4]	74.7	84.7	71.8
LoRAT [21]	72.1	81.8	70.7
ARTrackV2 [1]	75.9	85.4	72.7
ODTrack [40]	77.0	87.9	75.1
HIPTTrack [3]	77.4	88.0	74.5
UNTrack (Ours)	77.3	88.4	74.7

the GOT10K dataset [8], which is commonly used for RGB-based general tracking, providing a more comprehensive analysis. As shown in Tab. 4, UNTrack achieves performance comparable to other state-of-the-art trackers, slightly falling short on partial metrics. This is primarily due to the limited spectral information provided by RGB images, which restricts the performance of the spectrum prompt encoder. However, the asymmetric attention structure still enables robust feature extraction, demonstrating UNTrack’s potential across different data modalities.

5.4. Ablation Study

To perform a thorough analysis of the innovations introduced by UNTrack, we have conducted rigorous ablation studies on the proposed MUST dataset. All experiments adopt uniform training settings, as detailed in Sec. 5.1, to ensure consistency and isolate variables.

Number of Search Frames. UNTrack processes sequential searches instead of conventional single-frame search, thereby enabling unified feature extraction. As depicted in Tab. 5, we evaluate the influence of varying search frame

Table 5. Ablation study on the number of sequential search frames.

Number	AUC	SR _{0.5}	Pre	FLOPs	FPS
1	54.6	70.2	73.6	74.1G	54.6
2	59.7	75.8	79.2	106.9G	38.0
3	59.9	75.8	79.3	139.7G	20.5
4	58.5	73.9	76.1	172.5G	12.6
5	56.3	71.2	72.8	205.3G	9.2

Table 6. Comparison of different attention structures. SA and CA refer to self-attention and cross-attention applied to the prompt (P), template (T), and search (S). SC represents the introduced spectral background elimination mechanism.

SA			CA			SC	AUC	FLOPs	FPS
P	T	S	P	T	S				
–	✓	✓	–	✓	✓	–	53.8	140.9G	23.9
✓	✓	✓	✓	✓	✓	–	55.4	169.8G	21.4
✓	✓	✓	–	–	✓	–	59.5	137.5G	24.3
✓	✓	✓	–	–	✓	✓	59.7	106.9G	38.0

numbers on tracking performance. When transitioning from single-frame to sequence-based input, the tracker experiences a notable increase in AUC by 5.1%. However, further frame increments do not consistently translate into performance gains, and the computational burden is escalating. UAV tracking often involves drastic changes in the target state between consecutive frames, making excessive search frames likely to introduce learning burdens and potentially noise. Therefore, we made a trade-off between accuracy and efficiency, setting the number of search frames to 2.

Asymmetric Structure Configuration. To analyze the proposed asymmetric transformer, we compare different structural configurations, as shown in Tab. 6. Obviously, the introduction of spectrum prompt effectively improves AUC by 1.6%, while slowing down inference speed. We thus further introduce an asymmetric attention structure to prune cross-attention for both the prompt and template. The structural innovation greatly improves AUC by 4.1% while reducing FLOPs by 19.0%. Moreover, we compare the tracking performance before and after enabling the spectral background eliminate mechanism, which improves inference efficiency by progressively eliminating background while preserving tracking accuracy.

Spectrum Prompt Source. UNTrack employs a spectrum prompt encoder responsible for recording and updating the target spectral characteristics. We compare different approaches of generating spectrum prompt, as depicted in Tab. 7. In the absence of a spectrum prompt, UNTrack maintains a compact model size but suffers from suboptimal tracking accuracy. The prompt from random initialization without updating not only fails to enhance performance but

Table 7. Comparison of spectrum prompt from different sources.

Source	AUC	#Params	FPS
None	53.9	93.1M	39.9
Random Initialization	53.7	101.3M	39.2
Asymmetric Transformer	55.0	101.3M	39.2
Spectrum Prompt Encoder	59.7	112.6M	38.0

also introduces side effects by introducing irrelevant information. Notably, leveraging historical prompt for updates markedly boosts performance, and the encoded spectrum prompts exhibit stronger target representation capabilities.

Spectral Encoder. The proposed spectral encoder aims to generate prompts for subsequent tracking, which utilizes historical spectral features to improve performance. As shown Sec. 5.4, we first implement a Naive encoder, which improves AUC by 3.7% using only one convolution layer, proving the effectiveness of introducing a spectral encoder into network design. Notably, the Naive encoder struggles to generate robust prompts without spectral channels interaction. Therefore, we propose the Spectrum Prompt Encoder, which performs a compression-excitation operation on the spectral channels through FC and MLP layers. Our proposed encoder structure fully exploits spectral features and further improves AUC by 2.1%.

Encoder	AUC	SR _{0.5}	SR _{0.75}	Pre	Pre _N
w/o	53.9	68.1	41.5	71.9	67.7
Naive	57.6	72.4	45.3	76.6	70.8
Ours	59.7	75.8	46.2	79.2	74.8

6. Conclusion

This paper presented the first large-scale dataset for Multispectral UAV Single Object Tracking (MUST) and introduced UNTrack, a novel tracking framework that integrates spectral, spatial, and temporal features. UNTrack employs an asymmetric transformer with a spectral background eliminate mechanism and continuous spectrum prompt updates, achieving significant improvements in both tracking accuracy and efficiency. We believe the MUST dataset and the UNTrack framework will catalyze future advancements in UAV tracking research.

Limitations and Future Work Due to the principles and cost limitations of multispectral imaging, obtaining high spectral resolution video sequences at high frame rates is not trivial. In future work, we will collect video sequences with additional spectral bands and higher frame rates to address more challenging scenarios.

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